Cohort Dummy-Variable Extension to L: A Bayesian Linear Mixed Model Framework for SSBMF

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Abstract

When Supervised Sparse Bayesian Matrix Factorization (SSBMF) is trained on merged multi-platform cohorts (e.g., TCGA RNA-seq and CPTAC proteomics), the dominant principal component reflects platform-level technical variation rather than biology. The result is a collapsed survival linear predictor ($\beta \to 0$) that renders prognostic scoring useless. The standard workaround — per-platform z-score normalization before pooling — reduces but does not eliminate the problem and discards cohort-specific information.

This document proposes an explicit Bayesian linear mixed model (LMM) extension to the SSBMF L-matrix: appending C cohort indicator columns to L so that the genomics reconstruction absorbs platform-specific mean shifts through a new factor matrix F_{cohort}, while the survival linear predictor is protected by fixing $\beta_{\text{cohort}} \equiv 0$. The augmented model $Y = L_{\text{global}}F'_{\text{global}} + L_{\text{cohort}}F'_{\text{cohort}} + E$ retains closed-form CAVI updates for all components, requiring only modest modifications to the existing modular update pipeline.

This document is organized in three parts: (1) a literature review of Bayesian LMMs with emphasis on prior specifications and batch-effect corrections in genomics; (2) a detailed implementation proposal covering the model extension, CAVI update changes, prior choices, and API design; and (3) a concrete execution procedure with module-order implementation plan, test suite additions, synthetic and real-data validation criteria, rollback gates, and session-sized timeline. No model code is written or modified in this session; this document is the formal pre-implementation artifact.

1 Part 1: Bayesian Linear Mixed Models — Literature Review

1.1 Standard Model Formulation

The linear mixed model (LMM) is the foundational framework for jointly modelling fixed population-level effects and random subject- or group-level deviations. In its standard form,

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{Z}\mathbf{u} + \varepsilon, \qquad \mathbf{u} \sim \mathcal{N}(\mathbf{0}, \mathbf{G}), \quad \varepsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I}),$$

where $\mathbf{y} \in \mathbb{R}^n$ is the response vector, $\mathbf{X} \in \mathbb{R}^{n\times q}$ is the fixed-effects design matrix with coefficient vector $\beta \in \mathbb{R}^q$, $\mathbf{Z} \in \mathbb{R}^{n\times r}$ is the random-effects design matrix with random vector $\mathbf{u} \in \mathbb{R}^r$, and $\mathbf{G} = \sigma_u^2\mathbf{I}_r$ is the random-effect covariance. The key structural distinction is that β captures global, estimable contrasts of interest (e.g., treatment vs. control), while $\mathbf{u}$ absorbs group-specific deviations that are treated as exchangeable draws from a common population distribution.

Frequentist vs. Bayesian framing. In the frequentist tradition, the variance components (σ_u^2, σ^2) are estimated by restricted maximum likelihood (REML), which maximises the marginal likelihood after integrating out β. REML corrects for the degrees of freedom consumed by fixed effects and is unbiased for variance components in balanced designs (Chen and Dunson 2003). In a fully Bayesian treatment, both β and $\mathbf{u}$ receive prior distributions; what distinguishes $\mathbf{u}$ as *random* is the hierarchical prior with an *estimated* (or given a hyperprior) variance component σ_u^2, whereas β typically receives a fixed-scale weakly informative prior. The Bayesian posterior is

$$p(\beta, \mathbf{u}, \sigma_u^2, \sigma^2 \mid \mathbf{y}) \propto p(\mathbf{y} \mid \beta, \mathbf{u}, \sigma^2)\, p(\mathbf{u} \mid \sigma_u^2)\, p(\sigma_u^2)\, p(\beta)\, p(\sigma^2).$$

Full posterior inference is obtained via MCMC (Gibbs sampler for conjugate Normal–Inverse-Gamma models) or Variational Bayes (as in our SSBMF CAVI framework).

> **Bottom line.** The LMM separates signal (β) from structured noise ($\mathbf{u}$) through a hierarchical prior on $\mathbf{u}$. This separation is precisely what the cohort dummy-variable extension exploits: cohort membership enters $\mathbf{Z}$ (or equivalently, as additional columns of L) and the corresponding F_{cohort} plays the role of $\mathbf{u}$ — absorbing platform-specific mean shifts without contaminating the survival linear predictor.

1.2 Prior Specifications Used in Practice

The choice of prior on the random-effect coefficients and their variance components has substantial impact on inference, particularly in low-to-moderate sample size settings such as ours ($n \approx 150$–300 per cohort).

Normal priors with fixed scale are the default in probabilistic programming frameworks (`brms`, `rstanarm`). For a random effect $u \sim \mathcal{N}(0, \sigma_u^2)$, a half-Normal or weakly informative Normal prior is placed on σ_u. Gelman et al. (2008) proposed scaling all predictors to have mean 0 and standard deviation 0.5, then placing independent Cauchy(0, 2.5) priors on regression coefficients — a practical default that avoids both over-shrinkage of large effects and improper posteriors under separation.

These priors are computationally tractable and lead to conjugate Gibbs updates when the data model is Normal — directly relevant to the CAVI updates in `code/update_F.R`.

Half-Cauchy for variance components. Gelman (2006) critiqued Inverse-Gamma priors for variance parameters in hierarchical models on the grounds that they can be highly informative at small scales and recommended the half-Cauchy distribution as a default weakly informative prior for σ_u. The key property is that half-Cauchy$(0, s)$ has a heavy tail that permits large group-level variation while still regularizing near zero — important when some cohorts are small.

Sparsity-inducing priors. When the number of random effects is large relative to the sample size, shrinkage priors accelerate convergence and improve estimation:

- *Horseshoe* (Carvalho, Polson, and Scott 2010): places a half-Cauchy prior on each local shrinkage parameter λ_k and a global shrinkage τ, yielding a bimodal marginal that aggressively shrinks near-zero coefficients while leaving large signals intact. The horseshoe has been shown to be asymptotically minimax-rate optimal for sparse Normal means.
- *Regularized horseshoe* (Piironen and Vehtari 2017): addresses a practical limitation of the original horseshoe — the slab is infinitely wide, causing numerical instability for weakly identified parameters. The regularized variant introduces a finite slab width c^2, truncating the largest coefficients and improving posterior geometry for MCMC and variational methods.
- *Spike-and-slab* (George and McCulloch 1993): a two-component mixture with a point mass at zero (spike) and a diffuse Normal component (slab). This provides exact posterior sparsity and was historically the first Bayesian variable selection method to be computationally tractable via Gibbs sampling. It remains the conceptual foundation for continuous relaxations such as the horseshoe.

Empirical Bayes Normal Means (EBNM) and EBMF. Stephens (2017) introduced a flexible framework for the Normal means problem $\hat{\mu}_i = \mu_i + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, s_i^2)$, in which the prior on μ_i is estimated by marginal maximum likelihood from the data (empirical Bayes) rather than fixed in advance. The family of admissible priors includes Normal, point-Normal, point-Laplace, and non-parametric unimodal distributions. The `ebnm` R package (Willwerscheid, Carbonetto, and Stephens 2025) implements this framework efficiently. Wang and Stephens (2021) extended this to matrix factorization, introducing Empirical Bayes Matrix Factorization (EBMF): the model $Y = LF' + E$ is fit by a variational algorithm that reduces each factor update to an EBNM problem, estimating the appropriate amount of sparsity *from the data* rather than fixing a prior family in advance. SSBMF is a direct supervised extension of EBMF, adding a Cox partial likelihood on the linear predictor $\eta = L\beta$; the CAVI updates in `code/update_L.R`, `code/update_F.R`, and `code/update_beta.R` are all derived from the EBMF framework.

> **Bottom line.** For the cohort extension, the relevant prior choice is on F_{cohort}: a symmetric Normal prior (not point-Normal or point-Laplace) because cohort shifts can be positive or negative and are not expected to be sparse across genes. The Normal family is conjugate to the Normal genomics likelihood, giving a closed-form posterior — no EBNM optimisation loop required, just a scalar linear update.

1.3 Identifiability and Centering

Mixing fixed and random effects for the same grouping factor raises identifiability concerns that are particularly acute in small-C settings (our case: $C = 2$ for TCGA + CPTAC).

Centered vs. non-centered parameterizations. Betancourt and Girolami (2015) showed that for hierarchical models with small group sample sizes, the *centered* parameterization ($u_c \sim \mathcal{N}(0, \sigma_u^2)$, marginalised σ_u^2) creates a funnel geometry in the posterior that degrades MCMC mixing. The *non-centered* reparameterization ($u_c = \sigma_u \tilde{u}_c$, $\tilde{u}_c \sim \mathcal{N}(0, 1)$) decouples the scale from the direction and is preferred for small groups. In our CAVI setting the distinction is less critical — variational inference with a Normal approximation does not suffer from the HMC funnel problem — but the non-centered perspective is useful for understanding the Gaussian posterior update form.

Sum-to-zero vs. corner-point constraints. When L_{cohort} is a full rank $n \times C$ indicator matrix (one column per cohort, each entry $\in \{0, 1\}$), the column space contains the all-ones vector $\mathbf{1}_n$ if and only if every patient belongs to exactly one cohort. If Y is **column-centred** (zero gene mean across all patients), then $\mathbf{1}_n$ is projected out of Y by construction, and the augmented design matrix $[L_{\text{global}} \mid L_{\text{cohort}}]$ is not collinear with an intercept. This is precisely the case in SSBMF: `load_real_data()` applies `Y <- sweep(Y, 2, colMeans(Y), "-")` before fitting. With column-centred Y, corner-point identification (one reference cohort column dropped) is *not* required; all C indicator columns are included and their corresponding F_{cohort} rows are identified up to the prior.

Marginal vs. conditional gene means. Including all C cohort columns without a sum-to-zero constraint means that $F_{\text{cohort},c}$ estimates the *conditional* gene mean for cohort c given $L_{\text{global}} F'_{\text{global}}$, not a *contrast* against a reference cohort. This is the desired interpretation: each cohort's baseline expression level (after removing shared biology) is estimated independently, and their sum is naturally constrained by the column-centring of Y.

> **Bottom line.** With column-centred Y (the current SSBMF default), including all C cohort indicator columns is identifiable. No sum-to-zero constraint or corner-point parameterization is required, simplifying the implementation.

1.4 Batch Effects in Genomics via Mixed Models

The problem of unwanted technical variation in multi-cohort genomics data has a long methodological history. Four broad families of methods are relevant to SSBMF.

ComBat (empirical Bayes location-scale adjustment). Johnson, Li, and Rabinovic (2007) proposed ComBat, which models each gene's expression in batch b as $y_{ib} = \alpha_i + X_{ib}\beta_i + \gamma_{ib} + \delta_{ib}\varepsilon_{ib}$, where γ_{ib} is an additive batch shift and δ_{ib} a multiplicative scale factor. Empirical Bayes shrinkage of the batch parameters toward a common prior (estimated across genes) stabilizes estimates for genes with few samples per batch. ComBat has become the de-facto standard for pre-processing microarray and bulk RNA-seq data before downstream analysis. An extension for count data, ComBat-seq (Zhang, Parmigiani, and Johnson 2020), uses a negative-binomial regression model that preserves integer counts compatible with standard differential expression pipelines.

Surrogate Variable Analysis (SVA). Leek and Storey (2007) proposed a factor model for unmeasured confounders: estimate *surrogate variables* — latent factors capturing heterogeneity not explained by the primary variable of interest — by a two-step iterative SVD procedure, then include them as covariates in downstream models. SVA is designed for cases where batch labels are unknown or imprecise; it recovers latent structure rather than correcting known-batch effects.

Probabilistic Estimation of Expression Residuals (PEER). Stegle et al. (2010) introduced PEER, a Bayesian latent factor model for gene expression that explicitly accommodates *known* covariates alongside estimated latent factors. The expression of gene j in sample i is modelled as $y_{ij} = \sum_k \alpha_k X_{ik} + \sum_r \beta_r Z_{ir} + \varepsilon_{ij}$, where X contains known covariates (e.g., batch, platform, sex) with *fixed* indicator loadings and estimated gene-level weights α_k, and Z contains latent confounders with both loadings and weights estimated from data. This is the published precedent most directly analogous to the proposed cohort column extension: the known cohort indicator plays the role of X (fixed loading), the corresponding F_{cohort} row plays the role of α (estimated gene-level offset), and the global factors L and F play the role of the latent Z and β. PEER has been widely adopted in eQTL studies — including the GTEx consortium — as the standard method for controlling technical variation when batch labels are known. The key distinctions from the proposed approach are that PEER is unsupervised (no survival objective) and treats known-covariate estimation as a preprocessing step rather than integrating it jointly within CAVI.

Remove Unwanted Variation (RUV). Risso et al. (2014) proposed RUV-g and RUV-s, which use control genes (e.g., ERCC spike-ins) or replicate samples to estimate unwanted technical factors by factor analysis. The approach explicitly leverages negative control genes whose expression is known a priori to be unaffected by the biological factor of interest.

Supervised multi-omics integration. DIABLO (Singh et al. 2019) is a supervised multi-omics method based on sparse generalized canonical correlation analysis (sGCCA): it finds latent components that maximise cross-modality covariance while being predictive of a phenotype group. DIABLO is the most direct supervised competitor to SSBMF in the multi-omics literature, but it does not model batch membership, does not use a survival outcome (it targets categorical phenotypes), and has no principled way to exclude cohort effects from prediction — the latent components are free to exploit batch-level covariance if it correlates with the phenotype.

Matrix factorization approaches: MOFA, Harmony, and scVI. Multi-Omics Factor Analysis (Argelaguet et al. 2018) learns a shared low-dimensional representation across multiple omics layers using variational Bayes inference with ARD (spike-and-slab) priors — the architecturally closest method to SSBMF in the literature. MOFA decomposes each modality $Y^{(m)}$ as $LZ^{(m)\prime} + E^{(m)}$ where L is shared across modalities and $Z^{(m)}$ are modality-specific loadings; sparsity is induced on both. MOFA+ extends this to handle multiple sample groups, with separate $L^{(\text{group})}$ per batch (Argelaguet et al. 2020) — conceptually analogous to our L_{cohort} approach, but without a survival outcome and without an explicit $\beta_{\text{cohort}} = 0$ constraint to protect prognostic scores. Harmony (Korsunsky et al. 2019) iteratively adjusts PCA coordinates by soft-clustering cells and centering cluster centroids across batches. scVI (Lopez et al. 2018) uses a variational autoencoder with batch as a conditioning covariate, learning a latent code from which batch effects are explicitly decoded. Both operate in embedding space rather than correcting the raw data matrix.

Key distinction from SSBMF. The methods above treat batch correction as either a preprocessing step decoupled from downstream analysis (ComBat, SVA, RUV, Harmony), or as an unsupervised inference problem without an outcome (MOFA, MOFA+, scVI). None jointly optimize a survival objective while simultaneously separating batch from biological variation. The cohort dummy-variable extension proposed here performs **joint inference**: F_{cohort} is estimated within the CAVI iteration alongside the biological factors and the Cox partial likelihood, so the survival signal in L_{global} is continuously shielded from cohort contamination. The constraint $\beta_{\text{cohort}} \equiv 0$ makes this shielding exact and recoverable even when cohort effects dominate the first principal component — precisely the failure mode observed in merged TCGA + CPTAC training.

> **Bottom line.** PEER (Stegle et al. 2010) is the closest published architectural precedent: a Bayesian factor model that treats known covariates as fixed loadings with estimated gene-level magnitudes alongside latent factors — exactly the proposed cohort column structure. MOFA+ (Argelaguet et al. 2020) is the closest variational Bayes ancestor with group structure, but lacks a survival objective. ComBat (Johnson, Li, and Rabinovic 2007) is the closest analogue to F_{cohort} as a gene-mean correction, but operates as a preprocessing step. The proposed extension combines all three ideas — PEER's known-covariate architecture, MOFA+'s group-aware factor model, and ComBat's additive offset — within the SSBMF CAVI framework, with the addition of a Cox partial likelihood and the $\beta_{\text{cohort}} \equiv 0$ constraint.

1.5 Relevance to the SSBMF Setting

The literature review above motivates three design decisions that distinguish the proposed cohort extension from generic LMM implementations.

Why a Bayesian LMM framing fits SSBMF. SSBMF is a supervised extension of EBMF (Wang and Stephens 2021), which already places it within the Bayesian hierarchical framework: each factor update reduces to an EBNM problem, and the prior family (point-Normal, point-exponential, etc.) is estimated from the data. Adding cohort indicator columns to L with a *different* prior (Normal rather than point-Normal or point-exponential) is a natural extension of the same variational message-passing infrastructure — the update for F_{cohort} will be a closed-form Gaussian posterior rather than an EBNM optimisation, taking advantage of Normal-Normal conjugacy.

Why fixed (not random) cohort effects. With $C = 2$ to $C = 7$ cohorts in our PDAC data, the number of groups is small and there is no reason to *borrow strength* across cohorts via a common hyperprior on σ_u^2. In the random-effects sense, borrowing strength is most beneficial when $C \gg 1$ and many groups have few observations (Chen and Dunson 2003; Gelman 2006). With $C = 2$ (TCGA: $n_1 \approx 150$, CPTAC: $n_2 \approx 100$), both cohorts have ample data to estimate their own F_{cohort} row reliably; a shared hyperprior would add computational overhead without improving estimation. The prior variance σ_c^2 can simply be fixed (e.g., $\sigma_c^2 = 1$) or estimated as a single scalar EM-step on the Normal marginal likelihood, analogous to how EBNM estimates the prior scale.

Why $\beta_{\text{cohort}} = 0$ is principled, not a hack. The survival linear predictor $\eta_i = L_{\text{global},i}\beta_{\text{global}}$ should capture biological prognostic information — gene expression programs that predict patient outcome. Cohort membership is a *platform nuisance*: it encodes whether a patient was sequenced on an Illumina RNA-seq platform (TCGA) or profiled by mass spectrometry (CPTAC), not any intrinsic biology. Allowing $\beta_{\text{cohort}} \neq 0$ would permit the Cox partial likelihood to exploit cohort-level survival differences (which exist, if TCGA and CPTAC have different censoring rates or follow-up times), inflating the apparent C-index and producing a score that generalises poorly to new cohorts. Fixing $\beta_{\text{cohort}} = 0$ by construction enforces the interpretability and portability of the prognostic score.

> **Bottom line.** The cohort dummy-variable extension is a minimal, principled modification that fits naturally within the existing CAVI infrastructure. It adds $C \times p$ Normal posterior parameters (F_{cohort}), each with closed-form updates, and zero structural changes to the survival component. Section 2 specifies the model and all required CAVI modifications in detail.

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2 Part 2: Implementation Proposal

2.1 Model Specification

2.1.1 Current model (baseline)

The current SSBMF model jointly fits a genomics factor model and a Cox proportional hazards survival model:

$$Y_{n\times p} = L_{n\times K}\, F'_{K\times p} + E, \qquad E_{ij} \sim \mathcal{N}(0,\, \tau_j^{-1})$$

$$h_i(t) = h_0(t)\exp(\eta_i), \qquad \eta_i = \sum_{k=1}^{K} L_{ik}\,\beta_k = (L\beta)_i$$

Inference is by CAVI; each update reduces to an EBNM problem (Wang and Stephens 2021; Stephens 2017) for $q(L)$, $q(F)$, and $q(\beta)$, and to a closed-form precision MLE for $q(\tau)$.

2.1.2 Extended model (cohort dummy-variable)

Append C cohort indicator columns to L:

$$\underbrace{Y}_{n\times p} = \underbrace{L_{\text{global}}}_{n\times K}\, \underbrace{F'_{\text{global}}}_{K\times p} + \underbrace{L_{\text{cohort}}}_{n\times C}\, \underbrace{F'_{\text{cohort}}}_{C\times p} + E$$

where

- $L_{\text{cohort}} \in \{0,1\}^{n\times C}$ is the **fixed** cohort indicator matrix: $L_{\text{cohort},ic} = 1$ iff patient i belongs to cohort c, else 0. Each row has exactly one nonzero entry.
- $F_{\text{cohort}} \in \mathbb{R}^{C\times p}$ is **estimated**; row c is the cohort-c gene expression offset (positive or negative) that is absorbed by the genomics likelihood.

The survival linear predictor is **unchanged**:

$$\eta_i = (L_{\text{global}}\,\beta_{\text{global}})_i, \qquad \beta_{\text{cohort}} \equiv \mathbf{0}$$

Equivalently, stacking $[L_{\text{global}} \mid L_{\text{cohort}}]$ into an $n \times (K + C)$ augmented matrix and β as a $(K + C)$-vector fixed at zero in the last C positions recovers the same formulation in a notation consistent with `fit_modular.R`.

> **Bottom line.** The genomics likelihood sees all $K + C$ columns; the Cox partial likelihood sees only the first K. This is the entire structural change.

2.2 What Changes in the ELBO and CAVI Updates

Let $\mathbf{M} = L_{\text{cohort}} F'_{\text{cohort}}$ denote the cohort mean matrix ($n \times p$; row i equals $F_{\text{cohort,cohort}(i)}$).

$q(L_{\textsf{global}})$ — partial residual gains a cohort term

The partial residual R^{-k} currently computed by `compute_R_k` (`code/update_L.R:59`) is

$$R_{ij}^{-k} = Y_{ij} - \sum_{\ell \neq k} L_{i\ell} \bar{F}_{\ell j}$$

With the cohort extension it becomes

$$R_{ij}^{-k} = Y_{ij} - \sum_{\ell \neq k} L_{i\ell} \bar{F}_{\ell j} - \underbrace{F_{\text{cohort, cohort}(i),\, j}}_{\mathbf{M}_{ij}}$$

In practice this means subtracting the precomputed matrix $\mathbf{M}$ from Y before (or inside) the call to `compute_R_k`, or passing $Y - \mathbf{M}$ as the working response. The remainder of `update_L_k` (`code/update_L.R:122`) — the dual-source A_L, B_L construction and the EBNM call — is **unchanged**.

$q(L_{\textsf{cohort}})$ — point mass, no update

L_{cohort} is a fixed indicator matrix with zero posterior variance. No CAVI update is needed; it is constructed once from `cohort_id` before the iteration loop.

$q(F_{\textsf{global}})$ — same as current

`update_F_k` (`code/update_F.R:94`) computes

$$A_{F,j}^{(k)} = \tau_j \sum_i E[l_{ik}^2], \qquad B_{F,j}^{(k)} = \tau_j \sum_i R_{ij}^{-k} \bar{l}_{ik}$$

With the modified R^{-k} (cohort mean subtracted, as above), the computation is structurally identical.

$q(F_{\textsf{cohort}})$ — new closed-form Normal posterior

For cohort c and gene j, the genomics log-likelihood contribution (treating L_{cohort} as fixed) is

$$\sum_{i:\,\text{cohort}(i)=c} \frac{\tau_j}{2}\left(R_{\text{cohort},ij} - F_{\text{cohort},cj}\right)^2$$

where $R_{\text{cohort},ij} = Y_{ij} - (L_{\text{global}}\,\bar{F}'_{\text{global}})_{ij}$ is the full global-factor residual. Under a Normal prior $F_{\text{cohort},cj} \sim \mathcal{N}(0, \sigma_c^2)$, the posterior is Normal with natural parameters

$$\boxed{A_{\text{cohort},cj} = \tau_j\, n_c + \sigma_c^{-2}, \qquad B_{\text{cohort},cj} = \tau_j \sum_{i:\,\text{cohort}(i)=c} R_{\text{cohort},ij}}$$

Posterior mean and second moment:

$$\bar{F}_{\text{cohort},cj} = \frac{B_{cj}}{A_{cj}}, \qquad E[F^2_{\text{cohort},cj}] = A^{-1}_{cj} + \bar{F}^2_{\text{cohort},cj}$$

In matrix form (all cohorts and genes simultaneously):

```
R_cohort  <- Y - EL_global %*% t(EF_global)          # n x p
B_cohort  <- t(L_cohort) %*% R_cohort * rep_tau       # C x p  (element-wise)
A_cohort  <- n_c_vec %*% t(Tau) + 1/sigma_c^2         # C x p
EF_cohort <- B_cohort / A_cohort
EF2_cohort <- 1/A_cohort + EF_cohort^2
```

where `rep_tau` broadcasts τ across cohort rows, and `n_c_vec` is the C-vector of cohort sizes. This is $O(np+Cp)$ per iteration — negligible relative to the $O(npK)$ cost of the global factor updates.

q(tau) — Var_Term gains a cohort contribution

`compute_var_term` (`code/update_tau.R:60`) currently computes

$$\text{Var\_Term}_{ij} = \sum_k (E[l^2_{ik}]\, E[f^2_{jk}] - \bar{l}^2_{ik}\, \bar{f}^2_{jk})$$

Because L_{cohort} is a fixed indicator (zero posterior variance), $E[L^2_{\text{cohort},ic}] = L^2_{\text{cohort},ic}$ and the only new variance comes from F_{cohort}:

$$\text{Var\_Term}^{\text{cohort}}_{ij} = \sum_c L^2_{\text{cohort},ic}\, \text{Var}_q(F_{\text{cohort},cj}) = \text{Var}_q\Big(F_{\text{cohort},\,\text{cohort}(i),\,j}\Big) = A^{-1}_{\text{cohort},\,\text{cohort}(i),\,j}$$

In matrix form:

```
Var_cohort  <- 1 / A_cohort             # C x p
Var_cohort_n <- L_cohort %*% Var_cohort # n x p: broadcast to patients
```

This matrix is added to the existing `Var_Term` before computing the precision MLE in `update_tau` (`code/update_tau.R:86`).

ELBO — Normal KL term for $q(F_{\text{cohort}})$

`compute_ebnm_kl` (`code/compute_elbo.R:78`) handles KL terms for EBNM components. The cohort factors have an explicit Normal prior, so the KL is closed-form:

$$\mathrm{KL}[q(F_{\text{cohort},cj}) \,\|\, \mathcal{N}(0, \sigma_c^2)] = \frac{1}{2}\left(\frac{E[F^2_{\text{cohort},cj}]}{\sigma_c^2} + \log(A_{\text{cohort},cj}\,\sigma_c^2) - 1\right)$$

Summed over all c and j, this becomes the new `compute_normal_kl` function to be added to `code/compute_elbo.R`, integrated into `compute_total_elbo`.

> **Bottom line.** Five targeted modifications are needed: (1) subtract $\mathbf{M}$ from Y in `compute_R_k`; (2) add `update_F_cohort_all` as a new module; (3) extend `compute_var_term` to include cohort variance; (4) add `compute_normal_kl` to `compute_elbo.R`; (5) orchestrate in `fit_modular.R`. The $q(L_{\text{global}})$, $q(F_{\text{global}})$, and $q(\beta)$ update *forms* are unchanged.

2.3 Prior Choice for F_{cohort}

Recommended: $\mathcal{N}(0, \sigma_c^2)$ with shared fixed scale.

The Normal prior is preferred for three reasons:

1. **Sign symmetry.** A cohort may have systematically higher *or* lower expression than average for any gene. The one-sided priors used for F_{global} (point-exponential) are inappropriate here because they cannot represent negative cohort offsets.
2. **Conjugacy.** Normal prior + Normal likelihood = closed-form Normal posterior, avoiding the EBNM optimisation loop. This is computationally cheap and numerically stable.
3. **Column-centring compatibility.** With column-centred Y, the prior mean of zero is natural: the cohort offset is measured relative to the population gene mean (already removed), so $E[F_{\text{cohort},cj}] = 0$ a priori is correctly calibrated.

Scale choice. Two options:

Option	Formula	Pros	Cons
Fixed $\sigma_c^2 = 1$	Global default	No estimation step	May over- or under-shrink

Option	Formula	Pros	Cons
Empirical Bayes $\hat{\sigma}_c^2$	$\hat{\sigma}_c^2 = \frac{1}{p}\sum_j E[F_{cj}^2]$	Adapts to cohort effect size	One extra scalar EM step per cohort per iteration

The fixed default $\sigma_c^2 = 1$ is recommended for a first implementation. With column-centred, unit-variance-scaled gene expression, cohort offsets of order ± 1 standard deviation are the expected range for platform effects, so $\sigma_c^2 = 1$ is neither too tight nor too diffuse. The empirical Bayes option can be explored if fixed-scale shrinkage proves misspecified on real data (e.g., if CPTAC–TCGA offsets are consistently larger than 1 SD).

Alternative considered. A gene-specific prior σ_j^2 would allow some genes to have larger cohort variability than others. This is biologically motivated (housekeeping genes should have small cohort offsets; metabolic genes may have large ones) but introduces p additional parameters. Adding this as an optional `prior_F_cohort = "normal_genewise"` flag is left as future work and flagged as an open question for advisor review.

2.4 Why $\beta_{\text{cohort}} = 0$ Simplifies the Survival Update

Setting $\beta_{\text{cohort}} \equiv \mathbf{0}$ has three concrete consequences for the implementation:

1. `update_beta_all` loop length is unchanged. `update_beta_all` (`code/fit_modular.R:436`) loops over $k = 1, \ldots, K$. With $\beta_{\text{cohort}} = 0$ fixed, the loop still runs K iterations and returns a K-vector — no structural change.

2. Cox partial likelihood is unaffected. The working response z^{-k} and the Cox weights W_{ii} are computed from $\eta = L_{\text{global}}\beta_{\text{global}}$, which uses only the first K columns of the augmented L. No modification to `compute_z_no_k` or `compute_survival_elbo` (`code/compute_elbo.R:142`) is required.

3. Prognostic score portability. At prediction time (`code/predict.R`), the score for a new patient is $\hat{\eta} = \bar{L}_{\text{global}}\,\hat{\beta}_{\text{global}}$, projecting only onto the global factors. A new patient from an unseen cohort does not need a cohort label — the score is defined purely by their biology. This is the desired behaviour for external validation.

> **Bottom line.** $\beta_{\text{cohort}} = 0$ is not a constraint that requires special handling in the optimiser — it is simply the absence of a $q(\beta_{\text{cohort}})$ update step. Cohort membership is baked into L_{cohort}, but never allowed to reach the Cox likelihood.

2.5 Initialization Strategy

The key challenge is avoiding initialization collapse: if F_{cohort} starts at zero, the first CAVI pass for $q(L_{\text{global}})$ sees unresidualised Y and may assign the platform PC to $L_{\text{global},1}$ before F_{cohort} has a chance to absorb it.

Recommended two-pass initialization:

1. $F^{(0)}_{\textbf{cohort}}$**: per-cohort gene means** (closed-form, no iteration):

$$F^{(0)}_{\text{cohort},cj} = \frac{1}{n_c} \sum_{i:\,\text{cohort}(i)=c} Y_{ij}$$

   This is the ComBat-style starting point: absorb the raw per-cohort mean before factorisation begins. In R:

```
EF_cohort <- t(L_cohort) %*% Y / n_c_vec  # C x p
```

2. $Y^{(0)}_{\textbf{resid}}$**: cohort-mean-subtracted matrix**:

$$Y^{(0)}_{\text{resid},ij} = Y_{ij} - F^{(0)}_{\text{cohort},\text{cohort}(i),j}$$

   In R: `Y_resid <- Y - L_cohort %*% EF_cohort`

3. $L^{(0)}_{\textbf{global}}$, $F^{(0)}_{\textbf{global}}$**: SVD of** $Y^{(0)}_{\textbf{resid}}$, following the existing `svd` initialization in `fit_modular.R:248`.
4. $\beta^{(0)} = \mathbf{0}$: unchanged.

This ensures the platform PC is largely captured by $F^{(0)}_{\text{cohort}}$ before the first CAVI sweep, giving L_{global} a clean start on residual biological variation.

2.6 API Changes to `fit_modular.R`

The public interface of `fit_modular` gains three new arguments (all optional; defaults preserve existing behaviour exactly):

Argument	Type	Default	Description
`cohort_id`	character/factor, length n	`NULL`	Cohort membership vector; `NULL` disables extension
`prior_F_cohort`	character	`"normal"`	Prior family for F_{cohort} (currently only `"normal"` supported)
`sigma_F_cohort`	numeric scalar	`1.0`	Prior standard deviation σ_c (shared across cohorts and genes)

Internal logic (when `cohort_id` is non-NULL):

```
# Build fixed indicator matrix
L_cohort <- model.matrix(~ cohort_id - 1)   # n x C, no intercept
n_c      <- colSums(L_cohort)               # C-vector of cohort sizes
```

```r
# Initialise F_cohort (per-cohort gene means)
EF_cohort  <- t(L_cohort) %*% Y / n_c      # C x p
EF2_cohort <- EF_cohort^2 + sigma_F_cohort^2  # moment initialisation

# Residualised Y for SVD init
Y_init <- Y - L_cohort %*% EF_cohort

# CAVI loop: insert between L-update and F-update
M        <- L_cohort %*% EF_cohort           # n x p cohort mean
Y_adj    <- Y - M                            # adjusted working response
R_cohort <- Y_adj                            # global-factor residual
B_cohort <- (t(L_cohort) %*% R_cohort) *    # C x p
            matrix(Tau, C, p, byrow = TRUE)
A_cohort <- n_c %*% t(Tau) +                # C x p
            1 / sigma_F_cohort^2
EF_cohort  <- B_cohort / A_cohort
EF2_cohort <- 1/A_cohort + EF_cohort^2
```

Note: `Y_adj` *is recomputed each iteration as the global-factor estimates improve;* `R_cohort` *used above is the full global-factor residual (after subtracting all K global components), not just the partial k-residual.*

Return object gains three new fields:

```r
list(
  ...,                           # existing fields unchanged
  L_cohort    = L_cohort,        # n x C indicator (fixed)
  EF_cohort   = EF_cohort,       # C x p posterior means
  EF2_cohort  = EF2_cohort       # C x p posterior second moments
)
```

Backward compatibility: when `cohort_id = NULL`, every new code path is skipped entirely and the function produces bit-identical results to the current implementation. A regression test (`tests/test_fit_modular_cohort.R`) will verify this against a frozen ELBO trajectory.

2.7 Validation Plan

Validation proceeds in three stages of increasing realism.

Stage 1 — Synthetic data

Setup: $n = 200$, $p = 500$, $K_{\text{true}} = 3$, $C = 2$ balanced cohorts ($n_1 = n_2 = 100$).

- True $L_{\text{global}} \sim$ point_normal(0.7, 0, 1)
- True $F_{\text{global}} \sim$ point_exp(0.7, 1)

- True F_{cohort}: non-zero on first 100 genes only, $F_{\text{cohort},c,j} \sim \mathcal{N}(0,1)$; zero elsewhere
- True $\beta = (1.5, -1.0, 0.5)'$; Cox times from `simsurv`

Three models fit on the same data:

Label	Model
Baseline	`fit_modular(Y, cohort_id = NULL)`
Extended	`fit_modular(Y, cohort_id = cohort)`
Z-score	Per-cohort z-score + `fit_modular(Y_z, cohort_id = NULL)`

Acceptance criteria (all must pass to advance):

Criterion	Threshold
$\hat{\beta}$ recovery	$\text{cor}(\hat{\beta}, \beta_{\text{true}}) > 0.9$
F_{global} RMSE	Extended < Baseline
F_{cohort} recovery	$\text{cor}(\text{vec}(\hat{F}_{\text{cohort}}), \text{vec}(F_{\text{cohort,true}})) > 0.8$
Held-out C-index	Extended $\geq$ Baseline and Extended $\geq$ Z-score
ELBO monotonicity	Non-decreasing across all iterations

Stage 2 — Real TCGA + CPTAC merged

Run extended model with `cohort_id = c(rep("TCGA", n_tcga), rep("CPTAC", n_cptac))`. Compare to frozen `outputs/YFB_benchmark_perplatform/` numbers.

Acceptance criteria:

Criterion	Threshold
$\hat{\beta}$ non-collapse	≥ 2 of K components have $\lvert\hat{\beta}_k\rvert > 0.1$
External C-index	≥ 0.60 on ≥ 2 of 4 held-out cohorts
F_{cohort} separation	TCGA and CPTAC rows visually separate on PCA of $\hat{F}_{\text{cohort}}$

Stage 3 — All 7-cohort PDAC compendium

Extend to full 7-cohort PDAC dataset ($C = 7$): TCGA, CPTAC, Dijk, Moffitt, PACA-AU-seq, PACA-AU-array, Puleo. Evaluate factor stability via leave-one-cohort-out subsampling. No formal acceptance criteria at this stage — exploratory.

2.8 Open Questions and Risks

Eight issues are flagged for advisor discussion before implementation:

Q1 — Identifiability of $F_{\mathbf{cohort}}$ vs. $F_{\mathbf{global}}$ intercept. Column-centring of Y should suffice (see §1.3), but this requires verifying that `load_real_data()` applies centering before cohort indicators are appended. *Resolution: confirm centering in preprocessing pipeline; add assertion.*

Q2 — Prior scale σ_c^2 sensitivity. On real TCGA + CPTAC data, the platform offset may exceed 1 SD on many genes; an empirical Bayes $\hat{\sigma}_c^2$ update (single EM step per cohort) could be valuable. *Resolution: run fixed-σ_c^2 first; add EB option if Stage 2 shows systematic under-shrinkage.*

Q3 — Interaction with the α schedule. The mixing weight α in `update_L_k` (`code/update_L.R:87`) controls genomics vs. survival weighting. F_{cohort} enters only the genomics side ($\beta_{\text{cohort}} = 0$); it should therefore be computed from the fully-genomics residual, not the α-weighted partial residual. *Resolution: compute `R_cohort` from $Y - L_{global}\bar{F}'_{global}$ directly, bypassing α.*

Q4 — Sign correction compatibility. The existing post-fit sign correction (`code/fit_modular.R:590`) operates on $(L_{\text{global}}, F_{\text{global}}, \beta)$ columns. L_{cohort} is a fixed non-negative indicator, so the sign flip logic must explicitly skip cohort columns. *Resolution: add a guard `if (k <= K)` around the sign flip loop.*

Q5 — Prediction for unseen cohorts. At test time, a new patient from a cohort not in the training set has no L_{cohort} column. Two options: (a) project using only L_{global} (ignore cohort entirely), or (b) estimate $F_{\text{cohort}}^{\text{new}}$ from the test patient's residual and use it for reconstruction but not prediction. Option (a) is recommended for the prognostic score; option (b) may be useful for reconstruction quality diagnostics.

Q6 — Unbalanced cohorts. Small cohorts (e.g., Puleo, $n_c \approx 50$) will have higher posterior variance on $F_{\text{cohort},c}$; the Normal prior will shrink them more than large cohorts. This is desirable behaviour, but should be verified empirically.

Q7 — Computational cost. The F_{cohort} update is $O(np)$ per iteration (dominated by `t(L_cohort) %*% R_cohort`). With $K = 5$ global factors each requiring an $O(np)$ EBNM solve, the overhead is roughly $C/K \approx 40\%$ for $C = 2$. Acceptable for $n \leq 500$, $p \leq 2000$.

Q8 — Gene-specific prior σ_j^2. Allowing heterogeneous gene-level shrinkage would let the model learn that housekeeping genes have small platform offsets while metabolic or platform-specific genes have large ones. This adds p parameters but may improve F_{cohort} recovery. Flagged for advisor discussion; deferred to post-Stage-2 evaluation.

3 Part 3: Execution Procedure

This section translates the conceptual proposal in Part 2 into a concrete, step-numbered procedure. Each stage has defined acceptance criteria; failure at any gate blocks advancement to the next stage.

3.1 Derivation Document (Prerequisite, No Code)

Before any R code is written, produce `derivations/qF_cohort/qF_cohort_update_derivation.tex`, matching the structure of `derivations/qF/qF_update_derivation.tex`.

Required content:

1. Augmented model written out with explicit indices (i, j, c, k).
2. Variational factorization assumption, including the new factor $q(F_{\text{cohort}})$ as a product of independent Normal distributions over (c, j).
3. Derivation of $A_{\text{cohort},cj}$ and $B_{\text{cohort},cj}$ under a $\mathcal{N}(0, \sigma_c^2)$ prior, showing all algebra from the ELBO gradient to the closed-form posterior.
4. KL divergence term $\text{KL}[q(F_{\text{cohort},cj}) \| \mathcal{N}(0, \sigma_c^2)]$ and its contribution to the total ELBO.
5. Cohort contribution to the Var_Term in $q(\tau)$, showing why L_{cohort} has zero posterior variance and the simplification to $A^{-1}_{\text{cohort},cj}$.
6. Sign/identifiability discussion: why column-centred Y renders the all-cohorts parameterization identifiable (no sum-to-zero constraint required; see §1.3).

Acceptance criterion: PDF compiles via `pdflatex × 3` without errors; advisor mathematical review signs off before Session B begins.

3.2 Code Implementation — Module Order

Implement modular updates in strict dependency order. Each module follows the established pattern: one R source file + one test file + one demo script + one Quarto companion doc (see `MEMORY.md`, Modular Update Pattern). No module proceeds to the next until its tests pass.

2a. `code/update_F_cohort.R` — new file

Two exported functions:

```
update_F_cohort_k(cohort_k_idx, R_cohort, Tau, sigma_c, ...)
```

- `cohort_k_idx`: integer vector of row indices for cohort c
- `R_cohort`: $n \times p$ global-factor residual (precomputed)
- Returns `EF_cohort_k` (p-vector of posterior means) and `EF2_cohort_k` (p-vector of posterior second moments)

```
update_F_cohort_all(L_cohort, R_cohort, Tau, sigma_c)
```

- Loops over $c = 1, \dots, C$, calling `update_F_cohort_k`
- Returns `EF_cohort` ($C \times p$) and `EF2_cohort` ($C \times p$)

Closed-form update (see §2.2 for derivation):

```
B <- colSums(R_cohort[cohort_k_idx, ]) * Tau     # p-vector
A <- length(cohort_k_idx) * Tau + 1/sigma_c^2   # p-vector
EF_cohort_k  <- B / A
EF2_cohort_k <- 1/A + EF_cohort_k^2
```

2b. `code/update_L.R` — modify `compute_R_k` (line 59)

Add optional arguments `EF_cohort = NULL` and `L_cohort = NULL` to `compute_R_k`. When non-NULL, subtract the cohort mean matrix before returning:

```
compute_R_k <- function(Y, EL, EF, k, L_cohort = NULL, EF_cohort = NULL) {
  R <- Y - EL[, -k, drop = FALSE] %*% t(EF[, -k, drop = FALSE])
  if (!is.null(L_cohort))
    R <- R - L_cohort %*% EF_cohort
  R
}
```

Default values of `NULL` preserve existing caller behaviour exactly — no changes required to `update_L_all` (`code/update_L.R:267`) or any test relying on the current two-argument signature.

2c. `code/update_F.R` — pass cohort args to `compute_R_k`

`update_F_all` (`code/update_F.R:201`) calls `compute_R_k` internally. Thread the new `L_cohort` / `EF_cohort` arguments through from `update_F_all`'s signature to its `compute_R_k` call:

```
update_F_all <- function(Y, EL, EL2, EF, EF2, Tau,
                         L_cohort = NULL, EF_cohort = NULL, ...) {
  ...
  R_k <- compute_R_k(Y, EL, EF, k, L_cohort, EF_cohort)
  ...
}
```

Again, default `NULL` preserves backward compatibility.

2d. `code/update_tau.R` — extend `compute_var_term` (line 60)

```
compute_var_term <- function(EL, EL2, EF, EF2,
                             L_cohort = NULL, A_cohort = NULL) {
  base <- (EL2 %*% t(EF2)) - (EL^2 %*% t(EF^2))
  if (!is.null(L_cohort))
    base <- base + L_cohort %*% (1/A_cohort)   # n x p cohort variance term
  base
}
```

`A_cohort` ($C \times p$) is the posterior precision matrix from `update_F_cohort_all`; $1/A_{\text{cohort}}$ is the per-cell posterior variance. Broadcasting via `L_cohort %*% (1/A_cohort)` maps each patient to their cohort's variance row ($O(np)$ sparse matmul).

2e. `code/compute_elbo.R` — add `compute_normal_kl`

New function:

```
compute_normal_kl <- function(EF_cohort, EF2_cohort, sigma_c) {
  # KL[ N(mu, v) || N(0, sigma^2) ] = 0.5*(EF2/sigma^2 + log(A*sigma^2) - 1)
  # where A = 1/v (posterior precision) and EF2 = v + mu^2
  A <- 1 / (EF2_cohort - EF_cohort^2)          # C x p posterior precision
  kl_matrix <- 0.5 * (EF2_cohort / sigma_c^2
                      + log(A * sigma_c^2) - 1)
  sum(kl_matrix)                                # scalar
}
```

Integrate this into `compute_total_elbo` (when `EF_cohort` is non-NULL) as an additive KL penalty term.

2f. `code/fit_modular.R` — orchestration

Six targeted changes to the existing canonical CAVI loop:

Step	Location	Change
New args	Function signature	Add `cohort_id`, `prior_F_cohort`, `sigma_F_cohort`
Build indicator	Pre-iteration	`L_cohort <- model.matrix(~ cohort_id - 1)`
Init	Pre-iteration	Two-pass init: per-cohort means → residualised SVD

Step	Location	Change
F_cohort update	Inside k-loop, after L-update	Call `update_F_cohort_all(L_cohort, R_cohort, Tau, sigma_F_cohort)`
Return	Return list	Append `L_cohort`, `EF_cohort`, `EF2_cohort`
Backward compat	All new blocks	Guard with `if (!is.null(cohort_id))`

The `update_F_cohort_all` call is placed **between** the $q(L)$ sweep and the $q(F_{\text{global}})$ sweep: after L_{global} is updated but before F_{global} is refreshed, so both updates see the most recent estimate of the other. The precomputed global-factor residual $R_{\text{cohort}} = Y - \bar{L}_{\text{global}}\bar{F}'_{\text{global}}$ is passed directly (bypassing the α weighting, per §2.8 Q3).

3.3 Test Suite Additions

`tests/test_update_F_cohort.R` (~15 tests)

#	Test	Checks
T1	Closed-form correctness, balanced 2-cohort 3-gene toy	$\hat{F}_{cj} = B/A$ to machine precision
T2	Posterior variance correct	$E[F^2] - E[F]^2 = 1/A$
T3	Symmetric shifts cancel under tight prior	$\sigma_c^2 \to 0$: posterior mean $\to 0$
T4	Large data limit	$n_c \to \infty$: posterior mean $\to$ sample mean
T5	Prior dominates under weak data	$n_c = 1$, tight prior: shrinkage toward zero
T6	Single cohort equals intercept	$C = 1$: $\hat{F}_{1j} = \bar{Y}_j$ (column mean)
T7	Output dimensions	`EF_cohort` is $C \times p$; `EF2_cohort` same
T8	`EF2 >= EF^2` everywhere	Posterior second moment $\geq$ squared mean
T9	`update_F_cohort_all` agrees with manual loop	Vectorized vs. scalar loop
T10	Zero residual input	All residuals zero: posterior mean $\to 0$
T11	Different `sigma_c` values	Larger prior variance $\to$ less shrinkage
T12	Large p correctness	$p = 500$: no dimension mismatch
T13	Non-negative `EF2_cohort`	All entries strictly positive
T14	Tau scaling	Doubling `Tau` halves posterior variance
T15	`cohort_k_idx` respects patient ordering	Permuting patients gives same answer

`tests/test_fit_modular_cohort.R` (~10 tests)

#	Test	Checks
T1	`cohort_id = NULL` reproduces current results	ELBO trajectory matches frozen baseline to tolerance 10^{-6}
T2	Single-cohort `cohort_id = rep("A", n)` converges	No error; ELBO non-decreasing
T3	2-cohort synthetic: `EF_cohort` recovers known shifts	cor > 0.8 against ground truth
T4	`L_cohort` columns equal indicator matrix	`all(L_cohort %in% c(0, 1))` and row sums = 1
T5	`beta` stays length K	`length(fit$EBeta) == K` regardless of C
T6	ELBO is non-decreasing	Across all iterations for synthetic data
T7	Return object has new fields	`$L_cohort`, `$EF_cohort`, `$EF2_cohort` present
T8	`EF_cohort` rows sum near zero	Column-centred Y: cohort offsets cancel
T9	Var_Term augmented correctly	`sum(Var_Term_cohort) > 0` when cohort effects present
T10	`sigma_F_cohort` argument respected	Tighter prior produces smaller `mean(abs(EF_cohort))`

Run command:

```
Rscript tests/run_tests.R
```

Target: 193 (existing) + ~25 (new) = ~**218 tests passing**.

3.4 Demo Scripts

`demos/demo_update_F_cohort.R` — five scenarios following the established narrative pattern (see `demos/demo_update_F.R` for style):

Scenario	Setup	Illustrates
1	Single cohort ($C = 1$)	Degenerates to column-mean correction
2	Two balanced cohorts with known shift	Point-by-point recovery of F_{cohort}
3	Unbalanced cohorts ($n_1 = 200$, $n_2 = 20$)	Differential shrinkage; small cohort pulled toward prior
4	Prior strength sweep ($\sigma_c^2 \in \{0.1, 1, 10\}$)	Effect of prior specification on posterior mean magnitude
5	Real TCGA + CPTAC subset ($n = 50$ per cohort)	Illustrative application; not a validation

3.5 Documentation

Three documentation artifacts are required before the implementation is considered complete:

1. **`docs/update_F_cohort.qmd`** — Companion document following the established pattern (`docs/update_F.qmd` as template). Sections: code walkthrough, math-to-code mapping, test descriptions, demo walkthroughs. Rendered to PDF (xelatex) and HTML (cosmo) via `make -C docs all`.

2. **`code/SupervisedMF_Context.md`** — Add new symbols (L_{cohort}, F_{cohort}, A_{cohort}, B_{cohort}, $\mathbf{M}$) and update the CAVI call sequence diagram to show where `update_F_cohort_all` sits in the loop.

3. **`CLAUDE.md` quick-reference table** — Add rows for `code/update_F_cohort.R`, `tests/test_update_F_coho` `tests/test_fit_modular_cohort.R`, and `demos/demo_update_F_cohort.R`.

3.6 Verification Procedure — Synthetic Validation (Stage 1)

Create `results/cohort_lmm_sim/run_synthetic.R`.

Data-generating process:

```r
set.seed(42)
n <- 200; p <- 500; K_true <- 3; C <- 2
n_c <- c(100, 100)
cohort <- rep(1:C, n_c)
L_cohort <- model.matrix(~ factor(cohort) - 1)  # 200 x 2

# Global factors
L_global <- matrix(rbinom(n*K_true, 1, 0.7) * rnorm(n*K_true), n, K_true)
F_global <- matrix(rbinom(p*K_true, 1, 0.7) * rexp(p*K_true), p, K_true)

# Cohort shifts: non-zero on first 100 genes only
F_cohort_true <- matrix(0, C, p)
F_cohort_true[, 1:100] <- matrix(rnorm(C * 100), C, 100)

# Assemble Y
E <- matrix(rnorm(n * p, sd = 0.5), n, p)
Y <- L_global %*% t(F_global) + L_cohort %*% F_cohort_true + E
Y <- sweep(Y, 2, colMeans(Y), "-")  # column-centre

# True survival times (simsurv)
beta_true <- c(1.5, -1.0, 0.5)
eta_true  <- L_global %*% beta_true
# ... simulate Cox times with simsurv::simsurv(...)
```

Three models:

```r
fit_base <- fit_modular(Y, w, z, K = 3)
fit_ext  <- fit_modular(Y, w, z, K = 3, cohort_id = factor(cohort))
Y_z      <- scale_per_cohort(Y, cohort)  # per-cohort z-score
fit_z    <- fit_modular(Y_z, w, z, K = 3)
```

Acceptance criteria table (all must pass to advance to Stage 2):

Criterion	Model	Threshold	Pass/Fail
$\hat{\beta}$ recovery	Extended	$\text{cor}(\hat{\beta}, \beta_{\text{true}}) > 0.9$	TBD
F_{global} RMSE	Extended vs. Baseline	Extended RMSE < Baseline RMSE	TBD
F_{cohort} recovery	Extended	$\text{cor}(\text{vec}(\hat{F}_{\text{cohort}}), \text{vec}(F_{\text{cohort,true}})) > 0.8$	TBD
Held-out C-index	Extended vs. Baseline	Extended $\geq$ Baseline	TBD
Held-out C-index	Extended vs. Z-score	Extended $\geq$ Z-score	TBD
ELBO monotonicity	Extended	Non-decreasing all iterations	TBD

3.7 Verification Procedure — Real Data (Stage 2: TCGA + CPTAC)

Create `results/benchmark_sim/run_cohort_lmm_benchmark.R`, mirroring `run_LB_benchmark.R`.

Setup:

```r
# Load and merge TCGA_PAAD + CPTAC (existing load_real_data() pipeline)
cohort_id <- c(rep("TCGA", n_tcga), rep("CPTAC", n_cptac))

# Fit with alpha CV on log-lambda grid
fit <- fit_modular(Y_merged, w, z, K = K_selected,
                   cohort_id = cohort_id,
                   lambda = lambda_cv)

# External validation: project held-out cohorts using L_global factors only
C_ext <- sapply(ext_cohorts, function(cohort_data) {
  L_new  <- project_to_L(cohort_data$Y, fit)   # predict.R
  eta    <- L_new %*% fit$EBeta
  concordance(cohort_data$surv ~ eta)$concordance
})
```

Acceptance criteria:

Criterion	Threshold	Notes
$\hat{\beta}$ non-collapse	≥ 2 of K have $\lvert\hat{\beta}_k\rvert > 0.1$	vs. current merged collapse to ≈ 0
External C-index	≥ 0.60 on ≥ 2 of 4 held-out cohorts	Target: DeSurv range
F_{cohort} PC separation	TCGA and CPTAC visually separate on PC1-2	Visual diagnostic only
ELBO monotonicity	Non-decreasing	Sanity check

Comparison baseline: frozen `outputs/YFB_benchmark_perplatform/YFB_benchmark_results_merged.csv` and `outputs/LB_benchmark/LB_benchmark_results_merged.csv`.

3.8 Evaluation Procedure — Reporting

Write `docs/reports/ssbmf_cohort_lmm_results_MM_DD_YY.qmd` capturing:

1. **Synthetic validation tables** — $\hat{\beta}$ recovery correlation, F_{global} RMSE comparison, F_{cohort} correlation, C-index for all three models. One table per criterion.
2. **Real-data benchmark tables** — Per-cohort external C-index for Extended, LB-merged baseline, and z-score workaround. Training cohort C-index as a sanity check (should not be the primary metric).
3. **Loading heatmaps** — $\bar{L}_{\text{global}}$ columns ordered by $\lvert\hat{\beta}_k\rvert$; $\hat{F}_{\text{cohort}}$ rows (one per cohort) as a gene-wise barplot or heatmap.
4. **PCA of $\hat{F}_{\text{cohort}}$** — Visual diagnostic showing TCGA vs. CPTAC separation on the first two PCs of the cohort offset matrix.
5. **PH diagnostics** — Schoenfeld residual test for the cohort-adjusted $\eta = \bar{L}_{\text{global}}\hat{\beta}_{\text{global}}$; verify PH assumption holds after batch removal.
6. **Discussion** — When does the LMM extension help vs. when is preprocessing sufficient? Key variables: imbalance severity, cohort sample size, platform distance.

3.9 Rollback and Risk Gates

At each stage, defined exit criteria prevent bad results from propagating.

Stage	Gate condition	If fails $\rightarrow$ action
1 (Derivation)	PDF compiles; advisor sign-off	Pause; revise math before any code
2a (F_cohort module)	All 15 tests pass	Debug; do not proceed to 2b
2b–2e (Modifications)	Each module's existing tests still pass	No regressions allowed; fix before continuing
2f (fit_modular)	`cohort_id = NULL` reproduces frozen ELBO to 10^{-6}	Backward compat broken; must fix before synthetic run
3 (Synthetic Stage 1)	All 6 acceptance criteria pass	If $\hat{\beta}$ recovery < 0.9: debug update order / α interaction; do not advance to real data

Stage	Gate condition	If fails → action
4 (TCGA + CPTAC)	$\hat{\beta}$ non-collapse + C-index ≥ 0.60	If $\hat{\beta}$ still collapses: consider random-effect prior on σ_c^2 or stronger pre-centering
5 (7-cohort)	No numerical errors; ELBO converges	If unbalanced cohorts cause instability: revisit `sigma_F_cohort` or add cohort-specific prior

3.10 Timeline and Session Decomposition

The implementation is decomposed into five session-sized chunks. Each session corresponds to a single focused PR against `main`.

Session	Deliverable	Acceptance gate
A (1 session)	`derivations/qF_cohort/` — LaTeX derivation + PDF	Advisor mathematical review sign-off
B (1 session)	`code/update_F_cohort.R` + `tests/test_update_F_cohort.R` + `demos/demo_update_F_cohort.R` + `docs/update_F_cohort.qmd`	15/15 new tests pass
C (1 session)	Modifications to `compute_R_k`, `update_F_all`, `compute_var_term`, `compute_elbo.R`, `fit_modular.R` + `tests/test_fit_modular_cohort.R`	193 + ~25 = ~218 tests; backward-compat regression passes
D (1 session)	`results/cohort_lmm_sim/run_synthetic.R` — Stage 1 synthetic validation	All 6 Stage 1 acceptance criteria green
E (1–2 sessions)	`results/benchmark_sim/run_cohort_lmm_benchmark.R` — Stage 2 real-data + `docs/reports/ssbmf_cohort_lmm_results_MM_DD_YY.qmd`	Stage 2 acceptance criteria; report rendered

Sessions B and C together cover the full implementation of §2.2–2.6 and constitute the highest-risk work. Session A (derivation) is the prerequisite gate that must close before B can begin.

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Appendix A: Closed-Form Derivation of $q(F_{\textbf{cohort}})$

We derive the optimal variational factor $q^*(F_{\text{cohort},cj})$ under a mean-field factorization over (c, j).

Setup. The augmented genomics likelihood for a single gene j is

$$\log p(Y_{\cdot j} \mid L_{\text{global}}, F_{\text{global},\cdot j}, L_{\text{cohort}}, F_{\text{cohort},\cdot j}, \tau_j) = \frac{\tau_j}{2} \sum_i \left(Y_{ij} - \sum_k L_{ik} F_{kj} - \sum_c L_{\text{cohort},ic} F_{\text{cohort},cj} \right)^2 + \text{const}$$

For a fixed cohort c and gene j, the terms involving $F_{\text{cohort},cj}$ isolate to patients in cohort c (since $L_{\text{cohort},ic} = 0$ for cohort$(i) \neq c$). Define the global-factor residual for patient i:

$$R_{\text{cohort},ij} = Y_{ij} - \sum_k \bar{L}_{ik} \, \bar{F}_{kj}$$

The ELBO contribution involving $F_{\text{cohort},cj}$ is

$$\mathcal{L}_{cj} = \mathbb{E}_q \left[\frac{\tau_j}{2} \sum_{i \in c} \left(R_{\text{cohort},ij} - F_{\text{cohort},cj} \right)^2 \right] - \text{KL}[q(F_{\text{cohort},cj}) \| p(F_{\text{cohort},cj})]$$

Gradient w.r.t. $q(F_{\textbf{cohort},cj})$. Taking the functional derivative and setting to zero, the optimal q^* satisfies

$$\log q^*(F_{\text{cohort},cj}) \propto -\frac{1}{2} \left[\tau_j n_c + \sigma_c^{-2} \right] F^2_{\text{cohort},cj} + \left[\tau_j \sum_{i \in c} R_{\text{cohort},ij} \right] F_{\text{cohort},cj}$$

This is a Normal log-density in $F_{\text{cohort},cj}$ with natural parameters

$$A_{\text{cohort},cj} = \tau_j n_c + \sigma_c^{-2}, \qquad B_{\text{cohort},cj} = \tau_j \sum_{i \in c} R_{\text{cohort},ij}$$

so $q^*(F_{\text{cohort},cj}) = \mathcal{N}(\mu_{cj}, \, v_{cj})$ with

$$\mu_{cj} = \frac{B_{\text{cohort},cj}}{A_{\text{cohort},cj}}, \qquad v_{cj} = A^{-1}_{\text{cohort},cj}$$

Posterior second moment (needed for $q(\tau)$):

$$E_q[F^2_{\text{cohort},cj}] = v_{cj} + \mu^2_{cj} = A^{-1}_{\text{cohort},cj} + \mu^2_{cj}$$

KL divergence $\text{KL}[\mathcal{N}(\mu, v) \| \mathcal{N}(0, \sigma_c^2)]$:

$$\mathrm{KL}_{cj} = \frac{1}{2}\left(\frac{\mu_{cj}^2 + v_{cj}}{\sigma_c^2} + \log\frac{\sigma_c^2}{v_{cj}} - 1\right) = \frac{1}{2}\left(\frac{E_q[F_{\mathrm{cohort},cj}^2]}{\sigma_c^2} + \log(A_{\mathrm{cohort},cj}\,\sigma_c^2) - 1\right)$$

This is summed over all $c \in \{1, \dots, C\}$ and $j \in \{1, \dots, p\}$ to obtain the total cohort KL contribution to the ELBO.

Special case: $n_c \to \infty$. As $n_c \to \infty$, $A_{\mathrm{cohort},cj} \to \infty$ and the posterior concentrates: $\mu_{cj} \to \frac{1}{n_c}\sum_{i \in c} R_{\mathrm{cohort},ij}$ (the sample mean residual), $v_{cj} \to 0$. This is the MLE, confirming that the prior becomes irrelevant with sufficient data.

Special case: $\sigma_c^2 \to 0$. $A_{\mathrm{cohort},cj} \to \infty$ (prior dominates), $\mu_{cj} \to 0$, $v_{cj} \to 0$. The cohort effect is shrunk entirely to zero — equivalent to the no-cohort baseline.

Appendix B: Augmented CAVI Pseudocode

The following pseudocode shows the full CAVI loop for the extended model, with new cohort steps highlighted. Compare to `code/fit_modular.R` for the current implementation.

```
Input: Y (n x p), w (n-vector Cox weights), z (n-vector working response),
       K, cohort_id (n-vector or NULL), sigma_F_cohort

#   Pre-iteration setup
if cohort_id is not NULL:
    L_cohort  <- model.matrix(~ cohort_id - 1)          # n x C  [NEW]
    n_c       <- colSums(L_cohort)                       # C-vector [NEW]
    EF_cohort <- t(L_cohort) %*% Y / n_c                # C x p init [NEW]
    Y_for_svd <- Y - L_cohort %*% EF_cohort             # residualised [NEW]
else:
    Y_for_svd <- Y

EL, EF  <- svd_init(Y_for_svd, K)
EBeta   <- rep(0, K)
Tau     <- rep(1, p)
EL2, EF2, EBeta2 <- second_moments_from_init(EL, EF, EBeta)

#   CAVI iteration
repeat until convergence (delta_ELBO < tol):

    # Cohort mean matrix [NEW]
    if cohort_id is not NULL:
        M <- L_cohort %*% EF_cohort     # n x p

    # Step 1: Update q(beta) - unchanged
    for k = 1..K:
```

```
        z_no_k  <- compute_z_no_k(EL, EBeta, k)
        EBeta[k], EBeta2[k] <- update_beta_k(w, EL[,k], EL2[,k], z_no_k)

    # Step 2: Update q(L_global) - subtract cohort mean from Y [MODIFIED]
    for k = 1..K:
        Y_adj <- Y - M  (if cohort active, else Y)       # [NEW] subtract M
        R_k   <- compute_R_k(Y_adj, EL, EF, k)
        EL[,k], EL2[,k] <- update_L_k(Tau, EF[,k], EF2[,k], w,
                                        EBeta[k], EBeta2[k], R_k, z_no_k)

    # Step 3: Update q(F_cohort) - NEW block
    if cohort_id is not NULL:
        R_cohort <- Y - EL %*% t(EF)                     # global residual
        EF_cohort, EF2_cohort, A_cohort <-
            update_F_cohort_all(L_cohort, R_cohort, Tau, sigma_F_cohort)
        M <- L_cohort %*% EF_cohort                      # refresh M

    # Step 4: Update q(F_global) - uses updated M [MODIFIED]
    for k = 1..K:
        Y_adj <- Y - M  (if cohort active, else Y)       # [MODIFIED]
        R_k   <- compute_R_k(Y_adj, EL, EF, k)
        EF[,k], EF2[,k] <- update_F_k(Tau, EL[,k], EL2[,k], R_k)

    # Step 5: Update q(tau) - Var_Term augmented [MODIFIED]
    Var_cohort <- L_cohort %*% (1/A_cohort)  (if cohort active)
    EL2_R, EF2_R <- second_moments(EL, EL2, EF, EF2)
    Tau <- update_tau(Y - M, EL, EL2_R, EF, EF2_R,
                      extra_var = Var_cohort)

    # Step 6: ELBO (proxy) - add cohort KL [MODIFIED]
    ELBO <- compute_survival_elbo(...)
          + sum_k [compute_ebnm_kl(L_k) + compute_ebnm_kl(F_k)
                   + compute_ebnm_kl(beta_k)]
          + compute_normal_kl(EF_cohort, EF2_cohort, sigma_F_cohort)  # [NEW]
          + compute_tau_elbo(Tau, Y - M, EL, EL2_R, EF, EF2_R)

    check convergence

#   Post-iteration
apply sign_correction to (EL[,1..K], EF[,1..K], EBeta)  # cohort cols skipped

return list(EL, EL2, EF, EF2, EBeta, EBeta2, Tau,
            L_cohort, EF_cohort, EF2_cohort)             # [NEW fields]
```

Lines marked `[NEW]` are additions; lines marked `[MODIFIED]` are changes to existing logic. All unmarked lines are structurally unchanged from the current `fit_modular.R`. The total number of new lines in the CAVI loop is approximately 15–20, concentrated in Steps 3 and the cohort-mean subtractions in Steps 2 and 4.